# SHARP FROBENIUS ITERATION BOUND TURNING NON-SMOOTH INSEPARABLE POINTS INTO RATIONAL POINTS

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*Allowed standard prerequisites (editorial).*

- Original arithmetic geometry and number theory prerequisites. Original complete bibliography and source-local definitions retained.

## 1. Introduction

This paper is concerned with the behavior of non-smooth points on regular curves under iterated Frobenius pullbacks.

1.1. **Regularity and smoothness.** A point $x$ on a curve[1] $C$ over a field $K$ is called *regular* if its local ring $\mathcal{O}_{C,x}$ is regular. It is called *smooth* if it is regular and, in addition, the preimages of $x$ under the projection map $C_{\overline{K}} \to C$ are regular points on the extended curve $C_{\overline{K}} = C \otimes_K \overline{K}$ over the algebraic closure $\overline{K}$; see [EGA IV$_1$, Chapt. 0, 22.5.8], [Fal78], or [Liu02, p. 141].

If the base field $K$ is perfect (e.g., $\mathrm{char}(K) = 0$ or $K = \overline{K}$), regularity and smoothness are equivalent notions. If $K$ is imperfect, however, smoothness is stronger than regularity. A typical example of an imperfect field is the function field of an algebraic variety over a field of positive characteristic, such as $\mathbb{F}_q(t)$, where $t$ is transcendental over the finite field $\mathbb{F}_q$. In this context, regular curves that are non-smooth arise, making this phenomenon a distinctive feature of geometry in characteristic $p > 0$. For instance, *quasi-elliptic curves* (i.e., regular non-smooth curves of genus 1) occur as generic fibres of generically non-smooth fibrations on surfaces in characteristics $p = 2, 3$, which is the reason why the classification of algebraic surfaces is subtler in these characteristics than in other characteristics; see [BM77, BM76] or [CDL25, Chapter 4].

1.2. **Non-smooth points and Frobenius pullbacks.** Let $C$ be a regular curve over a field $K$ of characteristic $p > 0$. We consider the iterated Frobenius pullbacks $C^{(p^n)}$ of $C$, which fit into an infinite sequence of relative Frobenius morphisms

$$C^{(p^0)} = C \to C^{(p^1)} \to C^{(p^2)} \to C^{(p^3)} \to \cdots .$$

Passing to their normalizations $C_n$ we obtain a sequence of regular curves over $K$

$$C_0 = C \to C_1 \to C_2 \to C_3 \to \cdots .$$

Let $x$ be a non-smooth point on $C$. The non-smoothness of $x$ means that its geometric singularity degree $\delta(x) \geq 0$ is positive (see Definition 2.1). One knows that Frobenius pullbacks turn the non-smooth point $x$ into a smooth point, that is, the image point $x_n \in C_n$ is smooth for $n$ large enough. It was proved in [HS25] that the non-smooth

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[1]All our curves are assumed to be geometrically integral and projective.

point $x$ is actually turned into a $K$-rational point, if in addition $x$ is assumed to be *purely inseparable*, [2] i.e., $x \in C(K^{p^{-\infty}})$, where $K^{p^{-\infty}}$ is the maximal purely inseparable extension of $K$. Purely inseparable points, also known in the literature as *perfect*, have been studied in recent years by many authors [Ghi10, Rös15, Rös20, Yua21, BL22, Amb23].

**Definition 1.1.** Let $C|K$ be a regular curve equipped with a purely inseparable point $x$. We call $(C|K, x)$ a *pair*.

Given a pair $(C|K, x)$ with $x$ a non-smooth point, the bound $m$ such that $x_n \in C_n(K)$ for all $n \geq m$ depends on the singularity degree $d = \delta(x) > 0$ and the characteristic $p > 0$, but not on the curve $C$ nor on the field $K$ (see [HS25, Theorem 1.1]). In characteristic $p = 2$ the bound is sharp (see [HS25, Proposition 2.25]), that is, for each integer $d > 0$ the bound $m$ is optimal in the sense that there is a pair $(C|K, x)$ with $\delta(x) = d$ such that $x_{m-1} \notin C_{m-1}(K)$.

In characteristic $p > 2$ there is less flexibility, because on the one hand the singularity degrees $\delta(x)$ are multiples of $\frac{p-1}{2}$ (see [HS25, Proposition 2.5]), while on the other hand not every such multiple can occur as the singularity degree $\delta(x)$ of a pair $(C|K, x)$. We show (see Proposition 3.2) that a positive integer $d$ is *admissible*, i.e., there exists a pair $(C|K, x)$ with $\delta(x) = \frac{p-1}{2} \cdot d$, if and only if the congruence relation $d \not\equiv -1 \pmod{p}$ is satisfied. The goal of this paper is to determine a sharp bound for every admissible $d > 0$.

**Theorem 1.2.** *Let $(C|K, x)$ be a pair with $\delta(x) = \frac{p-1}{2} \cdot d > 0$. Then $x_n \in C_n$ is a $K$-rational point for all $n \geq \lambda_p(d)$, where $\lambda_p(d)$ is the integer determined in Theorem 3.5.*

*Furthermore the bound $\lambda_p(d)$ is sharp for $d > 0$, that is, there exists a pair $(C'|K', x')$ with $\delta(x') = \frac{p-1}{2} \cdot d$, such that $x'_n \in C'_n$ is not a $K'$-rational point for $n = \lambda_p(d) - 1$.*

One application of results like Theorem 1.2 is a method to achieve explicit descriptions of regular non-smooth curves $C|K$ (see [HS25, p. 281]). To explain how the method works, suppose $C$ is geometrically rational, i.e., the base change $C_{\overline{K}}$ has geometric genus $\overline{g} = h^1(\mathcal{O}_{\tilde{C}_{\overline{K}}}) = 0$, where $\tilde{C}_{\overline{K}}$ is the normalization of $C_{\overline{K}}$. This implies that the regular curve $C_n$ has genus 0 for large $n$. Thus if some non-smooth point $x$ on $C$ is purely inseparable, [3] the theorem yields an integer $n > 0$ such that $x_n$ is a rational point on the genus zero curve $C_n$, hence $C_n \cong \mathbb{P}^1(K)$. Now, starting with this explicit description of $C_n$ one can successively obtain explicit descriptions of the regular curves $C_{n-1}, C_{n-2}, \ldots, C$, by applying an algorithm due to Bedoya and Stöhr [BS87].

The above strategy was applied in [HS25b, Section 2] to get a full description of all quasi-elliptic curves. It has also been applied to obtain a complete classification of all regular geometrically rational plane quartic curves in characteristic two (see [HS24]).

**Remark 1.3.** The sharp bounds $\lambda_p(d)$, defined as the smallest $m$ such that for all pairs $(C|K, x)$ with $\delta(x) = \frac{p-1}{2} \cdot d$ one has $x_n \in C_n(K)$ for all $n \geq m$, depend on the collection of pairs $(C|K, x)$ considered, e.g., restricting the collection of pairs may lead to larger sharp bounds. In addition, the sharp bounds depend solely on the properties of the non-smooth points $x$, i.e., they are independent of the properties of the regular curves $C|K$ at their points $y \neq x$. It is reasonable to expect that the same sharp bounds $\lambda_p(d)$ will hold if

[2] In [HS25] one uses the term *non-decomposed*; see Definition 2.2.

[3] The existence of non-smooth purely inseparable points on $C$ can be assumed without loss of generality, by passing, if necessary, to the regular curve $C_L = C \otimes_K L$, where $L$ is the separable closure of $K$.

we impose suitable *global* restrictions on the pairs $(C|K, x)$, where by *global* we mean conditions not involving the point $x$. We check that this is the case for the restrictions $\overline{g} = h^1(\mathcal{O}_{\tilde{C}_{\overline{K}}}) = 0$ and $\operatorname{pdeg}(K) > 1$ (see Remarks 1.4 and 1.5 below). We expect the same to be true for other global restrictions such as requiring that $x$ be the only non-smooth point on $C$, or that the extended curve $C_{\overline{K}}$ have a fixed positive geometric genus $\overline{g}$.

1.3. **Outline of proof of the theorem.** To compute the sharp bounds $\lambda_p(d)$ we construct a class of pairs $(C|K, x)$ whose image points $x_n \in C_n$ are not $K$-rational for suitable $n$. This yields a lower bound for $\lambda_p(d)$ (we have an upper bound by [HS25, Theorem 1.1]), which, after a careful analysis, leads to the actual value of $\lambda_p(d)$. A key ingredient in our approach is the duality between regular curves and function fields (see [EGA II, 7.4], or [Sil09, II.2.5] for the case where the base field is perfect), which allows us to investigate the problem in the arithmetic setting of function field theory.

**Remark 1.4.** A by-product of our construction is a subclass of pairs $(C|K, x)$ that realize the sharp bounds $\lambda_p(d)$. While explicit descriptions of each point $x$ and its images $x_n$ are given, we have no a priori control over the remaining points on the curve $C$, hence it is hard to describe its global geometry, e.g., compute the genus $g = h^1(\mathcal{O}_C) = h^1(\mathcal{O}_{C_{\overline{K}}})$. These curves, however, share a common feature: they are all geometrically rational, i.e., each base change $C_{\overline{K}}$ has geometric genus $\overline{g} = h^1(\mathcal{O}_{\tilde{C}_{\overline{K}}}) = 0$.

**Remark 1.5.** Some of the pairs $(C|K, x)$ we construct exist only over imperfect base fields $K$ whose $p$-degree $\operatorname{pdeg}(K) = \log_p[K : K^p] > 0$ is at least 2 (see Remark 4.3). This implies that the same sharp bounds $\lambda_p(d)$ hold if we only consider pairs $(C|K, x)$ with $\operatorname{pdeg}(K) > 1$. It is not clear what happens when $\operatorname{pdeg}(K) = 1$.

1.4. **Arbitrary non-smooth points.** For points that are not purely inseparable our sharp bounds $\lambda_p(d)$ are related to the notion of a separable point. Given a point $x$ on a regular curve $C|K$, one knows (see [HS25, Corollary 2.16]) that the separable degree $[\kappa(x) : K]_s$ of the residue field extension $K \subset \kappa(x)$ divides the integer $\frac{2}{p-1} \cdot \delta(x)$.

**Theorem 1.6** (see Theorem 3.11)**.** *Let $C|K$ be a regular curve. Let $x$ be an (arbitrary) non-smooth point on $C$ with $\delta(x) = \frac{p-1}{2} \cdot [\kappa(x) : K]_s \cdot d > 0$. Then the image point $x_n \in C_n$ is separable, i.e., $x_n \in C_n(K^{sep})$, for all $n \geq \lambda_p(d)$.*

*Furthermore $\lambda_p(d)$ is sharp with respect to this property, that is, there exist a regular curve $C'|K'$ and a non-smooth point $x'$ on $C'$ with $\delta(x') = \frac{p-1}{2} \cdot [\kappa(x') : K]_s \cdot d > 0$, such that $x'_n \in C'_n$ is not separable for $n = \lambda_p(d) - 1$.*

The theorem generalizes Theorem 1.2, since a point $x$ is purely inseparable if and only if $[\kappa(x) : K]_s = 1$, and moreover the image point $x_n$ is rational if and only if it is both separable and purely inseparable.

## 2. Preliminaries

By a *curve* over a field $K$ we mean a proper geometrically integral $K$-scheme of dimension 1. A curve is called *regular* (or *normal*) if its local rings are regular local rings.

As follows from [EGA II, 7.4], there is a one-to-one contravariant correspondence between the regular curves $C$ over a field $K$ and the *one-dimensional separable function*

*fields* $F|K$, given by the assignment $C|K \mapsto F|K = K(C)|K$. A one-dimensional separable function field is a separable field extension $F|K$ of transcendence degree one, with $K$ algebraically closed in $F$. Note that if $K \subset K'$ is an algebraic extension then $FK'|K' = F \otimes_K K'|K'$ is also a one-dimensional separable function field; the corresponding regular curve over $K'$ is the normalization of the base change $C_{K'} = C \otimes_K K'$.

In the function field setting the (regular) closed points on the curve $C|K$ are called the *primes* of the function field $F|K$, and they are denoted by the letters $\mathfrak{p}$, $\mathfrak{q}$, etc. The local rings $\mathcal{O}_\mathfrak{p}$ of the primes $\mathfrak{p}$ of $F|K$ are the (discrete) valuation rings of $F|K$.

**Definition 2.1.** The *geometric singularity degree* $\delta(\mathfrak{p})$ of a prime $\mathfrak{p}$ is the $\overline{K}$-codimension of the semilocal domain $\mathcal{O}_\mathfrak{p} \otimes_K \overline{K} \subset F \otimes_K \overline{K}$ in its integral closure. Equivalently, let $\pi : \tilde{C}_{\overline{K}} \to C_{\overline{K}}$ be the normalization morphism of the extended curve $C_{\overline{K}} = C \otimes_K \overline{K}$ over $\overline{K}$; then $\delta(\mathfrak{p}) = \sum_y \delta(y)$, where $y$ runs through the points on $C_{\overline{K}}$ that lie over $\mathfrak{p} \in C$ and $\delta(y)$ is the $\overline{K}$-dimension of the stalk $(\pi_* \mathcal{O}_{\tilde{C}_{\overline{K}}} / \mathcal{O}_{C_{\overline{K}}})_y$.

A prime $\mathfrak{p}$ is called *singular* if $\delta(\mathfrak{p}) > 0$, i.e., $\mathfrak{p}$ is non-smooth as a point on $C|K$. Singular primes exist only over base fields $K$ that are imperfect, in particular only if the characteristic $p$ of $K$ is positive. We assume throughout that $p > 0$.

Let $F|K$ be a (one-dimensional separable) function field. The Frobenius pullback $C^{(p)}|K$ of the corresponding regular curve $C|K$ is a curve that may not be regular. Passing to its normalization we get a regular curve $C_1|K$ with function field $F_1|K := F^p K|K$. We call $F_1|K$ the *Frobenius pullback* of $F|K$. Iterating this process we get iterated Frobenius pullbacks $F_n|K = F^{p^n}K|K$, which are the function fields of the regular curves $C_n|K = \widetilde{C^{(p^n)}}|K$. The composite $F_n = F^{p^n}K$ is the only intermediate field of $F|K$ such that the extension $F_n \subset F$ is purely inseparable of degree $p^n$.

Given a prime $\mathfrak{p}$, there is an associated discrete valuation $v_\mathfrak{p}$ of $F|K$. Restricting $v_\mathfrak{p}$ to $F_n|K$ and normalizing we obtain a discrete valuation that corresponds to a prime $\mathfrak{p}_n$ of $F_n|K$. We call $\mathfrak{p}_n$ a *restricted prime*, or the *restriction* of $\mathfrak{p}$ to $F_n|K$.

By the Fundamental Equality for function fields, the product between the ramification and inertia indices $e_{\mathfrak{p}|\mathfrak{p}_1}$ and $f_{\mathfrak{p}|\mathfrak{p}_1}$ of each extension $\mathfrak{p}|\mathfrak{p}_1$ equals $[F : F_1] = p$. If $e_{\mathfrak{p}|\mathfrak{p}_1} = 1$, i.e., if the purely inseparable extension $\kappa(\mathfrak{p}_1) \subset \kappa(\mathfrak{p})$ has degree $p$, we say that $\mathfrak{p}|\mathfrak{p}_1$ is *unramified* or *inertial*; if $e_{\mathfrak{p}|\mathfrak{p}_1} = p$, i.e., $\kappa(\mathfrak{p}_1) = \kappa(\mathfrak{p})$, we say that $\mathfrak{p}|\mathfrak{p}_1$ is *ramified*.

By [HS25, Proposition 2.9], a prime $\mathfrak{p}$ is *separable*, i.e., the residue field extension $K \subset \kappa(\mathfrak{p})$ is separable, i.e., $\mathfrak{p} \in C(K^{sep})$, if and only if $\mathfrak{p}$ is non-singular and the extension $\mathfrak{p}|\mathfrak{p}_1$ is ramified. In particular singular primes do not exist if $K$ is perfect. Moreover, every prime $\mathfrak{p}$ that is rational, i.e., $\kappa(\mathfrak{p}) = K$, is also non-singular.

**Definition 2.2.** A prime $\mathfrak{p}$ is called *purely inseparable* (or *non-decomposed*) if it satisfies the following three equivalent conditions (see [HS25, Corollary 2.17])

(i) there is a unique prime in the extended function field $F\overline{K}|\overline{K}$ that lies over $\mathfrak{p}$, i.e., there is a unique point in the extended curve $C_{\overline{K}}$ that lies over $\mathfrak{p} \in C$;
(ii) the restricted prime $\mathfrak{p}_n$ is rational for some $n$;
(iii) the residue field extension $K \subset \kappa(\mathfrak{p})$ is purely inseparable, i.e., $\mathfrak{p} \in C(K^{p^{-\infty}})$.

Note that if some restricted prime $\mathfrak{p}_n$ is rational then so is each $\mathfrak{p}_m$ for $m \geq n$, since $K \subset \cdots \subset \kappa(\mathfrak{p}_2) \subset \kappa(\mathfrak{p}_1) \subset \kappa(\mathfrak{p})$. Clearly $\mathfrak{p}$ is purely inseparable if and only if some (and every) restricted prime $\mathfrak{p}_n$ is purely inseparable.

The local invariants (in particular the singularity degree) of any purely inseparable prime $\mathfrak{p}$ can be calculated from the Bedoya–Stöhr algorithm [BS87]. We remark that the key condition for the algorithm is item (ii) in Definition 2.2 (see [BS87, p. 313]), although in [BS87] one always takes for granted the stronger assumption that $K$ is separably closed.

**Theorem 2.3** ([BS87, Theorem 2.3])**.** *Let $\mathfrak{p}$ be a purely inseparable prime in a one-dimensional separable function field $F|K$. Choose $n > 0$ such that the restricted prime $\mathfrak{p}_n$ of $F_n|K$ is rational. Let $z \in F$ be a function such that $\mathcal{O}_\mathfrak{p} = \mathcal{O}_{\mathfrak{p}_1}[z]$. Then*

$$\delta(\mathfrak{p}) = p\,\delta(\mathfrak{p}_1) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_n}(dz^{p^n}).$$

Here $v_{\mathfrak{p}_n}(dz^{p^n})$ is the order of the exact differential $dz^{p^n}$ of $F_n|K$ at the rational prime $\mathfrak{p}_n$. Note that a function $z$ with $\mathcal{O}_\mathfrak{p} = \mathcal{O}_{\mathfrak{p}_1}[z]$ always exists: if the extension $\mathfrak{p}|\mathfrak{p}_1$ is ramified, one can take $z$ to be a local parameter at $\mathfrak{p}$; if $\mathfrak{p}|\mathfrak{p}_1$ is inertial one can select a function $z \in \mathcal{O}_\mathfrak{p}$ whose residue class $z(\mathfrak{p}) \in \kappa(\mathfrak{p})$ generates the degree $p$ purely inseparable extension $\kappa(\mathfrak{p}_1) \subset \kappa(\mathfrak{p})$.

The algorithm is applied backwards: starting with $\delta(\mathfrak{p}_n) = 0$, the theorem yields successively $\delta(\mathfrak{p}_{n-1}), \delta(\mathfrak{p}_{n-2}), \ldots, \delta(\mathfrak{p})$; see Section 4 for explicit applications.

**Corollary 2.4.** *Assumptions as in Theorem 2.3. Then $2\delta(\mathfrak{p}) \not\equiv 1 \pmod{p}$.*

*Proof.* Let $t \in F_n$ be a local parameter at the rational prime $\mathfrak{p}_n$. Write $z^{p^n} \in F_n$ as a Laurent series in $t$ with coefficients in $K$, say $z^{p^n} = \sum_i a_i t^i$. Then $dz^{p^n} = \sum_i i a_i t^{i-1} dt$, and since we work in characteristic $p > 0$ we conclude $v_{\mathfrak{p}_n}(dz^{p^n}) \not\equiv -1 \pmod{p}$. □

## 3. The sharp bound

We start by fixing some terminology and notation.

**Definition 3.1.** Let $F|K$ be a one-dimensional separable function field. Assume that there is a purely inseparable prime $\mathfrak{p}$ in $F|K$. We call $(F|K, \mathfrak{p})$ a *pair*.

By [HS25, Proposition 2.5], the geometric singularity degree $\delta(\mathfrak{p})$ of a pair $(F|K, \mathfrak{p})$ is a multiple of $\frac{p-1}{2}$, that is, $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot d$ for some integer $d$. We say that a positive integer $d$ is *admissible* if there exists a pair $(F|K, \mathfrak{p})$ such that $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot d$.

**Proposition 3.2.** *A positive integer $d$ is admissible if and only if $d \not\equiv -1 \pmod{p}$.*

*Proof.* Suppose that $d \not\equiv -1 \pmod{p}$. The function field $F|K = K(x, z)|K$ given by

$$z^p = a + x^{d+1}, \quad \text{where } a \in K \setminus K^p$$

has Frobenius pullback $F_1|K = K(x)|K$. Let $\mathfrak{p}$ be the zero of $x$, i.e., let $\mathfrak{p}$ be the only prime such that $v_\mathfrak{p}(x) > 0$. This prime is purely inseparable, since its restriction $\mathfrak{p}_1$ is the rational $x$-adic valuation of $F_1|K$. As the residue class $z(\mathfrak{p}) \in \kappa(\mathfrak{p})$, which equals $a^{1/p}$, does not belong to $\kappa(\mathfrak{p}_1) = K$ it follows that $\mathfrak{p}$ is inertial over $F_1$ (i.e., $\mathfrak{p}|\mathfrak{p}_1$ is inertial) with residue field $\kappa(\mathfrak{p}) = K(z(\mathfrak{p})) = K(a^{1/p})$. As the differential $dz^p = (d+1)x^d dx$ of $F_1|K$ has order $d$ at $\mathfrak{p}_1$, we conclude from Theorem 2.3 that

$$\delta(\mathfrak{p}) = p\,\delta(\mathfrak{p}_1) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_1}(dz^p) = \frac{p-1}{2} \cdot d.$$

Thus $d$ is admissible, as desired. The converse follows from Corollary 2.4. □

**Remark 3.3.** Let $F|K = K(x,z)|K$ be the function field in the above proof. Let $\mathfrak{p}'$ be the pole of $x$, i.e., $\mathfrak{p}'$ is the only prime with $v_{\mathfrak{p}'}(x) < 0$. By the Jacobian criterion every prime different from $\mathfrak{p}$ and $\mathfrak{p}'$ is non-singular. The prime $\mathfrak{p}'$ is also non-singular, actually rational, because $v_{\mathfrak{p}'_1}(z^p) = -(d+1)$ is not divisible by $p$ and so the extension $\mathfrak{p}'|\mathfrak{p}'_1$ is ramified. Thus $\mathfrak{p}$ is the only singular prime of $F|K$. Since the rational Frobenius pullback $F_1|K$ has genus $g_1 = 0$ the extended function field $F\overline{K}|\overline{K}$ has genus $\overline{g} = 0$, hence we conclude from [HS25, Formula 2.3] that the function field $F|K$ has genus $g = \frac{p-1}{2}\cdot d$. Note that for $p > 2$, $d = 1$, this recovers the curve considered in the proof of [IIL20, Lemma 8.13].

For each positive integer $d$ that is admissible we define

$$\tau_p(d) = \begin{cases} \lceil \log_p((p-1)d+1) \rceil & \text{if } d \text{ is a sum of consecutive } p\text{-powers,} \\ \lceil \log_p((p-1)d+1) \rceil - 1 & \text{otherwise,} \end{cases}$$

$$= \begin{cases} i+1 & \text{if } d = P_j^i \text{ for some } j \le i, \\ i & \text{if } P_0^{i-1} < d < P_0^i \text{ and } d \ne P_j^i \text{ for all } j \text{ with } j \le i, \end{cases}$$

where for every pair of integers $i \ge j \ge 0$ the symbol $P_j^i$ denotes

$$P_j^i = p^j + \cdots + p^i = \sum_{k=j}^{i} p^k = \frac{p^{i+1} - p^j}{p-1}.$$

**Theorem 3.4** ([HS25, Theorem 2.24])**.** *Let $(F|K, \mathfrak{p})$ be a pair such that $\delta(\mathfrak{p}) = \frac{p-1}{2}\cdot d > 0$. Then the restricted prime $\mathfrak{p}_n$ is rational for all $n \ge \tau_p(d)$.*

Our goal in this paper is to find a bound similar to $\tau_p(d)$ that is sharp in every characteristic $p > 0$. In other words, given an admissible integer $d > 0$, we ask for the smallest integer $m$ such that for every pair $(F|K, \mathfrak{p})$ with $\delta(\mathfrak{p}) = \frac{p-1}{2}\cdot d$ the restricted prime $\mathfrak{p}_n$ is rational for all $n \ge m$. We denote the smallest such $m$ by $\lambda_p(d)$. Now Theorem 3.4 rephrases as

$$1 \le \lambda_p(d) \le \tau_p(d). \tag{1}$$

The theorem below gives the precise value of $\lambda_p(d)$.

**Theorem 3.5.** *Let $d$ be a positive integer such that $d \not\equiv -1 \pmod p$. Choose $0 \le r < p-1$ such that $d \equiv r \pmod p$. Then $\lambda_p(d) = \tau_p(d)$ or $\lambda_p(d) = \tau_p(d) - 1$, the latter case occurring if and only if there is an integer $i > 1$ such that $P_0^{i-1} < d < (r+1)P_0^{i-1}$ and*

$$d \ne rP_0^{i-1} + P_j^{i-1} \quad \text{for all } j \text{ with } 0 < j \le i.$$

Note that for $i < j$ we follow the convention that $P_j^i = 0$. As an immediate consequence we retrieve [HS25, Proposition 2.25].

**Corollary 3.6.** *If a positive integer $d$ satisfies $d \equiv 0 \pmod p$ then the bound $\tau_p(d)$ is sharp for $d$, i.e., $\lambda_p(d) = \tau_p(d)$. In particular, in characteristic $p = 2$ the bound $\tau_p(d)$ is always sharp.*

The proof of the theorem requires some preparation. In view of (1), a first step consists in bounding $\lambda_p(d)$ from below. For this we need explicit examples of pairs $(F|K, \mathfrak{p})$ with the property that the restricted primes $\mathfrak{p}_n$ are not rational for suitable $n$. We postpone the discussion of such examples to the next section, and use them here to prove the theorem.

**Proposition 3.7.**

(i) *Let $i \geq j > 0$, $\ell \geq 0$ and $r \in \{0, \dots, p-2\}$ be integers. Assume that $d = rP_0^i + \ell P_j^i$ is positive, so in particular $d$ is admissible. Then $i < \lambda_p(d)$.*

(ii) *Let $i > j > 0$, $\ell \geq 0$ and $r, r' \in \{0, \dots, p-2\}$ be integers. Assume that $d = rP_0^i + P_j^i + r'p^j + \ell p^{j+1}$ is positive, so $d$ is admissible. Then $i < \lambda_p(d)$.*

The proposition follows immediately from Examples 4.1 and 4.2. It provides lower bounds for $\lambda_p(d)$ in specific cases. We use these specific bounds to derive a general bound for $\lambda_p(d)$.

**Lemma 3.8.** *Let $d$ be a positive integer such that $d \equiv r \pmod p$, where $0 \leq r < p-1$. Suppose that $rP_0^i + P_1^i - 2p < d$ for some integer $i > 0$. Then $i < \lambda_p(d)$.*

*Proof.* In view of Proposition 3.7, it is enough to show that $d$ can be written as

$$d = rP_0^i + \ell p^i \quad \text{for some } \ell \geq 0, \tag{2}$$

or as

$$d = rP_0^i + P_j^i + r'p^j + \ell p^{j+1} \text{ for some } \ell \geq 0,\ r' \in \{0, \dots, p-2\} \text{ and } j \text{ with } 0 < j < i. \tag{3}$$

Note first that $-p^i \leq P_1^i - 2p < d - rP_0^i$. Moreover for each $r' \in \{0, \dots, p-2\}$ and $j$ with $0 < j < i$ one has

$$-p^{j+1} \leq P_j^i - 2p^j - P_j^i - r'p^j \leq P_1^i - 2p - P_j^i - r'p^j < d - rP_0^i - P_j^i - r'p^j.$$

Thus, if $d$ does not admit a representation as in (3), i.e., $d - rP_0^i - P_j^i - r'p^j \not\equiv 0 \pmod{p^{j+1}}$ for each $r' \in \{0, \dots, p-2\}$ and $j$ with $0 < j < i$, then

$$d - rP_0^i \not\equiv (r'+1)p^j \pmod{p^{j+1}} \quad \text{for each } j = 0, \dots, i-1 \text{ and } r' = 0, \dots, p-2,$$

which means that $d - rP_0^i \equiv 0 \pmod{p^i}$, i.e., $d$ admits a representation as in (2). □

Given a prime $\mathfrak{p}$, recall (see [HS25, pp. 284-285]) that the sequence of singularity degrees $\delta(\mathfrak{p}_n)$ is non-decreasing and converges to zero rapidly in the sense that the partial differences

$$\Delta_n = \Delta_n(\mathfrak{p}) := \delta(\mathfrak{p}_n) - \delta(\mathfrak{p}_{n+1}) \geq 0$$

satisfy $\Delta_{n+1} \leq p^{-1}\Delta_n$. In turn $\delta(\mathfrak{p}) = \Delta_0 + \Delta_1 + \Delta_2 + \cdots$. Moreover each $\Delta_n$ is a multiple of $\frac{p-1}{2}$.

**Lemma 3.9.** *Let $(F|K, \mathfrak{p})$ be a pair. Let $n > 0$ be such that $\mathfrak{p}_n$ is rational. Assume that*

$$\Delta_{i-1}(\mathfrak{p}) = \frac{p-1}{2} \cdot rp^{n-i} \quad \text{for every } i = 1, \dots, n,$$

*where $0 < r < p-1$. Then the extension $\mathfrak{p}|\mathfrak{p}_n$ is totally inertial, i.e., $[\kappa(\mathfrak{p}) : \kappa(\mathfrak{p}_n)] = [F : F_n] = p^n$. In particular the degree $\deg(\mathfrak{p}) = [\kappa(\mathfrak{p}) : K]$ of $\mathfrak{p}$ is equal to $p^n$.*

*Proof.* We apply descending induction. The extension $\mathfrak{p}_{n-1}|\mathfrak{p}_n$ is inertial because $\mathfrak{p}_n$ is rational and $\delta(\mathfrak{p}_{n-1}) = \Delta_{n-1}(\mathfrak{p}) > 0$. Suppose that $\mathfrak{p}_i|\mathfrak{p}_n$ is totally inertial for some $i < n$, and let us check that $\mathfrak{p}_{i-1}|\mathfrak{p}_i$ is inertial. If this were false, then [HS25b, Lemma 3.2] would imply $\delta(\mathfrak{p}_{i-1}) - p\,\delta(\mathfrak{p}_i) \geq \frac{p-1}{2} \cdot \deg(\mathfrak{p}_i)$, i.e., $r \geq p^{n-i}$, a contradiction. □

**Remark 3.10.** According to [HS25] the integer $\tau_p(d)$ admits the combinatorial description

$$\tau_p(d) = \max\{s + \min\{v_p(d_1), \dots, v_p(d_s)\}\}, \tag{4}$$

where $v_p$ is the $p$-adic valuation of $\mathbb{Q}$ and the maximum is taken over all the partitions

$$d = d_1 + \cdots + d_s$$

of $d$ such that

$$d_{i+1} \leq p^{-1} d_i \quad \text{for each } i = 1, \dots, s-1.$$

Every pair $(F|K, \mathfrak{p})$ with $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot d$ yields a partition $d = d_1 + \cdots + d_s$ of $d$, where $d_i = \frac{2}{p-1} \cdot \Delta_{i-1}(\mathfrak{p})$ and $s$ is the smallest integer such that $\mathfrak{p}_s$ is non-singular. By the proof of [HS25, Proposition 2.22], in this case the restricted prime $\mathfrak{p}_n$ is rational for all $n \geq s + \min\{v_p(d_1), \dots, v_p(d_s)\}$. Therefore, the bound $m = \tau_p(d)$ in Theorem 3.4 is sharp for $d$, i.e., $\lambda_p(d) = \tau_p(d)$, if and only if the maximum in (4) is attained by a partition $d = d_1 + \cdots + d_s$ that comes from a pair $(F|K, \mathfrak{p})$ with the property that $\mathfrak{p}_{m-1}$ is non-rational.

Now we have all the ingredients needed to prove Theorem 3.5.

*Proof of Theorem 3.5.* Recall that $1 \leq \lambda_p(d) \leq \tau_p(d)$. If $d = P_j^i$ for some $i \geq j \geq 0$ then the bound $\tau_p(d) = i+1$ is sharp for $d$, i.e., $\lambda_p(d) = \tau_p(d)$, as follows from Proposition 3.7 (i). Thus we may assume that there is an integer $i > 0$ such that the following holds

$$P_0^{i-1} < d < P_0^i \quad \text{and} \quad d \neq P_j^i \text{ for all } j \leq i.$$

We wish to see when the bound $\tau_p(d) = i$ is sharp for $d$. As the bound is sharp when $\tau_p(d) = 1$ we can further assume $i > 1$. In addition, since the bound is sharp when $(r+1)P_0^{i-1} \leq d < P_0^i$, as this implies by Lemma 3.8 that $i - 1 < \lambda_p(d)$, we may suppose that

$$P_0^{i-1} < d < (r+1)P_0^{i-1},$$

and in turn that $r > 0$, $p > 2$. Therefore, we must prove that in this situation

(i) the bound $\tau_p(d) = i$ is sharp if and only if $d$ can be written as $d = rP_0^{i-1} + P_j^{i-1}$ for some $j$ with $0 < j \leq i$;
(ii) if the bound $\tau_p(d) = i$ is not sharp then $\lambda_p(d) = \tau_p(d) - 1$.

We prove (i). As the if part follows from Proposition 3.7 (i) we just need to show the only if part. For any partition $d = d_1 + \cdots + d_s$ of $d$ we have $s + \min\{v_p(d_1), \dots, v_p(d_s)\} = s$, since $r > 0$, hence in light of Remark 3.10 the condition that the bound $\tau_p(d) = i$ is sharp means that there exist a partition $d = d_1 + \cdots + d_i$ of length $i$ and a pair $(F|K, \mathfrak{p})$ such that

$$\mathfrak{p}_i \text{ is rational} \quad \text{and} \quad \Delta_{k-1}(\mathfrak{p}) = \frac{p-1}{2} \cdot d_k \text{ for all } k = 1, \dots, i.$$

It is clear that

$$p^{i-k} \leq d_k < p \cdot p^{i-k} \quad \text{for all } k = 1, \dots, i,$$

since $d_{k+1} \leq p^{-1} d_k$ for each $k$ and $d_1 < p^i$ as $(r+1)P_0^{i-1} \leq (p-1)P_0^{i-1} < p^i$. As $\mathfrak{p}_i$ is rational and $\delta(\mathfrak{p}_{i-1}) = \Delta_{i-1}(\mathfrak{p}) > 0$ we deduce that $\mathfrak{p}_{i-1}$ has degree $p$ and hence that $p$ divides $d_k$ for every $k < i$ (see [HS25, Proposition 2.8]). Therefore $d_i = r$, so in particular

$$rp^{i-k} \leq d_k < p \cdot p^{i-k} \quad \text{for all } k = 1, \dots, i.$$

Choose a positive integer $j \le i$ that is maximal with respect to the following property

$$d_k = rp^{i-k} \quad \text{for all } k > i-j.$$

By Lemma 3.9, the prime $\mathfrak{p}_{i-j}$ has degree $p^j$, which implies that $p^j$ divides $d_k$ for each $k \le i-j$ (see [HS25, Proposition 2.8]). Now $(r+1)p^j \le d_{i-j}$, and hence $(r+1)p^{i-k} \le d_k$ for each $k \le i-j$, so we conclude that the integer

$$d - (rP_0^{i-1} + P_j^{i-1}) = \sum_{1 \le k \le i-j} \big(d_k - (r+1)p^{i-k}\big)$$

is a non-negative multiple of $p^j$. In view of $d < (r+1)P_0^{i-1}$, this means $d = rP_0^{i-1} + P_j^{i-1}$.

It remains to prove (ii). Since $1 \le \lambda_p(d) < \tau_p(d)$ we may assume $\tau_p(d) > 2$, i.e., $i > 2$. As $P_0^{i-1} < d$ we conclude from Lemma 3.8 that $i - 2 < \lambda_p(d)$, i.e., $\tau_p(d) - 2 < \lambda_p(d)$. □

A prime $\mathfrak{p}$ is said to be *separable* if the residue field extension $K \subset \kappa(\mathfrak{p})$ is separable. Recall that the separable degree $[\kappa(\mathfrak{p}) : K]_s$ of $K \subset \kappa(\mathfrak{p})$ divides the integer $\frac{2}{p-1}\delta(\mathfrak{p})$; see [HS25, Corollary 2.16].

**Theorem 3.11.** *Let $F|K$ be a one-dimensional separable function field. Let $\mathfrak{p}$ be a (not necessarily purely inseparable) singular prime with $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot [\kappa(\mathfrak{p}) : K]_s \cdot d > 0$. Then the restricted prime $\mathfrak{p}_n$ is separable for all $n \ge \lambda_p(d)$.*

*Proof.* Let $\mathfrak{q}$ be a (purely inseparable) prime above $\mathfrak{p}$ in the extended function field $FL|L$, where $L = K^{sep}$. Its restrictions $\mathfrak{q}_n$ lie above the restrictions $\mathfrak{p}_n$ of $\mathfrak{p}$. By [HS25, Proposition 2.12] the prime $\mathfrak{q}$ has $\delta(\mathfrak{q}) = \delta(\mathfrak{p})/[\kappa(\mathfrak{p}) : K]_s$ and $\deg(\mathfrak{q}) = \deg(\mathfrak{p})/[\kappa(\mathfrak{p}) : K]_s$. Analogous equalities hold for the restricted primes $\mathfrak{p}_n$ and $\mathfrak{q}_n$. Thus $\mathfrak{p}_n$ is separable if and only if $\mathfrak{q}_n$ is rational. □

The proof shows that the integers $\lambda_p(d)$ are sharp bounds in the following sense. Let $d$ be a positive integer that is admissible. Then $\lambda_p(d)$ is the smallest integer $m$ that satisfies the following property: given a one-dimensional separable function field $F|K$ and a prime $\mathfrak{p}$ in $F|K$ with $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot [\kappa(\mathfrak{p}) : K]_s \cdot d$, the restricted prime $\mathfrak{p}_n$ is separable for all $n \ge m$.

## 4. Examples

This section discusses the examples of function fields and purely inseparable primes that were needed in the previous section to bound $\lambda_p(d)$ from below. To compute singularity degrees we systematically use the Bedoya–Stöhr algorithm (see Theorem 2.3).

**Example 4.1.** Let $i \ge j > 0$, $r \in \{0, \dots, p-2\}$ and $\ell \ge 0$. We construct a one-dimensional separable function field $F|K$ having a purely inseparable prime $\mathfrak{p}$ such that

$$\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot \big(rP_0^i + \ell P_j^i\big), \quad \mathfrak{p}_i \text{ is non-rational}, \quad \mathfrak{p}_{i+1} \text{ is rational}.$$

Let $K$ be an imperfect field containing elements

$$a \in K \setminus K^p \quad \text{and} \quad b \in K \setminus K^p(a^{1/p^j}). \tag{5}$$

For instance, $K$ can be the function field of the projective plane $\mathbb{P}^2(k)$ over an algebraically closed ground field $k$, or over a finite field $k = \mathbb{F}_q$. Consider the function field $F|K = K(x,y)|K$ defined by the equation

$$y^{p^{i+1}} = b^{p^j} + x^{\ell \cdot p^j}(a + x^{r+1}).$$

Set $z := x^{-\ell}(b + y^{p^{i-j+1}})$, so that $z^{p^j} = a + x^{r+1}$ and $y^{p^{i-j+1}} = b + x^\ell z$. The Frobenius pullbacks of $F|K$ are given by

$$F_n|K = \begin{cases} K(x, z, y^{p^n})|K, & \text{if } 0 \le n \le i-j, \\ K(x, z^{p^{n-i+j-1}})|K, & \text{if } i-j < n < i+1, \\ K(x)|K, & \text{if } n = i+1. \end{cases}$$

Let $\mathfrak{p}$ be the zero of the function $x$, i.e., $v_{\mathfrak{p}}(x) > 0$. Its restriction $\mathfrak{p}_{i+1}$ to the rational function field $F_{i+1}|K = K(x)|K$ is the rational $x$-adic valuation. In particular, $\mathfrak{p}$ is purely inseparable.

We compute the singularity degree $\delta(\mathfrak{p})$, and on the way check that $\mathfrak{p}_i$ is non-rational. As is clear from $z(\mathfrak{p})^{p^j} = a \notin K^p = \kappa(\mathfrak{p}_{i+1})^p$, for every $n$ with $i-j < n < i+1$ the extension $\mathfrak{p}_n|\mathfrak{p}_{n+1}$ is unramified, or more precisely, the residue field extension $\kappa(\mathfrak{p}_{n+1}) \subset \kappa(\mathfrak{p}_n)$ is purely inseparable of degree $p$, generated by the residue class $z(\mathfrak{p})^{p^{n-i+j-1}}$ of the function $z^{p^{n-i+j-1}} \in F_n$. As the differential $dz^{p^j} = (r+1)x^r dx$ of $F_{i+1}|K = K(x)|K$ has order $r$ at $\mathfrak{p}_{i+1}$, from Theorem 2.3 we infer

$$\delta(\mathfrak{p}_n) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_{i+1}}(dz^{p^j}) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot r, \quad (i-j < n < i+1).$$

Similarly, since

$$y(\mathfrak{p})^{p^{i-j+1}} = \begin{cases} b, & \text{if } \ell > 0, \\ b + a^{1/p^j}, & \text{if } \ell = 0, \end{cases}$$

does not lie in $\kappa(\mathfrak{p}_{i-j+1})^p = K^p(a^{1/p^{j-1}})$, for every $n \le i-j$ the extension $\mathfrak{p}_n|\mathfrak{p}_{n+1}$ is unramified and

$$\delta(\mathfrak{p}_n) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_{i+1}}(dy^{p^{i+1}}) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot (r + \ell p^j), \quad (0 \le n \le i-j),$$

where the last equality is due to the fact that the differential $dy^{p^{i+1}} = x^{\ell p^j} dz^{p^j}$ of $F_{i+1}|K$ has order $\ell p^j + r$ at $\mathfrak{p}_{i+1}$. Thus the $\Delta$-invariants of the singular prime $\mathfrak{p}$ are equal to

$$\Delta_n(\mathfrak{p}) = \delta(\mathfrak{p}_n) - \delta(\mathfrak{p}_{n+1}) = \begin{cases} \frac{p-1}{2} \cdot (r+\ell)p^{i-n}, & \text{if } 0 \le n \le i-j, \\ \frac{p-1}{2} \cdot r p^{i-n}, & \text{if } i-j < n \le i, \\ 0, & \text{if } i < n. \end{cases}$$

In particular $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot (rP_0^i + \ell P_j^i)$, as required. Moreover,

$$\deg(\mathfrak{p}_n) = \begin{cases} p^{i+1-n}, & \text{if } 0 \le n \le i+1, \\ 1, & \text{if } i+1 \le n. \end{cases}$$

**Example 4.2.** Let $i > j > 0$, $r, r' \in \{0, \dots, p-2\}$ and $\ell \ge 0$. We construct a function field $F|K$ and a purely inseparable prime $\mathfrak{p}$ such that

$$\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot \left(rP_0^i + P_j^i + r'p^j + \ell p^{j+1}\right), \quad \mathfrak{p}_i \text{ is non-rational}, \quad \mathfrak{p}_{i+1} \text{ is rational}.$$

Take an imperfect field $K$ and consider the function field $F|K = K(y,u)|K$ given by the relation

$$z^{p^j+r+1} = az^{r+1} + y^{p^{i-j}(r+1)},$$

where $z := u^p - y^{p\ell+r'+1}$ and $a \in K \setminus K^p$. Let $x := y^{p^{i-j}}/z$, so that

$$z^{p^j} = a + x^{r+1}, \quad y^{p^{i-j}} = xz, \quad u^p = z + y^{p\ell+r'+1}.$$

Then the Frobenius pullbacks take the form

$$F_n|K = \begin{cases} K(x,z,y,u)|K, & \text{if } n = 0, \\ K(x,z,y^{p^{n-1}})|K, & \text{if } 1 \le n \le i-j, \\ K(x,z^{p^{n-i+j-1}})|K, & \text{if } i-j < n < i+1, \\ K(x)|K, & \text{if } n = i+1. \end{cases}$$

Let $\mathfrak{p}$ be the (purely inseparable) zero of the function $x$. Its restriction $\mathfrak{p}_{i+1}$ to the rational function field $F_{i+1}|K = K(x)|K$ is a rational prime with local parameter $x$. As follows from $z(\mathfrak{p})^{p^j} = a \notin K^p = \kappa(\mathfrak{p}_{i+1})^p$, the extension $\mathfrak{p}_{i-j+1}|\mathfrak{p}_{i+1}$ is unramified and

$$\delta(\mathfrak{p}_n) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_{i+1}}(dz^{p^j}) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot r, \quad (i-j < n < i+1).$$

Since $y^{p^{i-j}} = xz \in F_{i-j+1}$ is a local parameter at $\mathfrak{p}_{i-j+1}$ the extension $\mathfrak{p}_1|\mathfrak{p}_{i-j+1}$ is totally ramified, that is, for every $n$ with $1 \le n \le i-j$ the extension $\mathfrak{p}_n|\mathfrak{p}_{n+1}$ is ramified and $y^{p^{n-1}}$ is a local parameter at $\mathfrak{p}_n$. Using Theorem 2.3 we deduce

$$\delta(\mathfrak{p}_n) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_{i+1}}(dy^{p^i}) = p\,\delta(\mathfrak{p}_{n+1}) + \frac{p-1}{2} \cdot (p^j + r), \quad (1 \le n \le i-j).$$

Now, because $u(\mathfrak{p}) = z(\mathfrak{p})^{1/p}$ does not lie in $\kappa(\mathfrak{p}_1) = \kappa(\mathfrak{p}_{i-j+1}) = K(z(\mathfrak{p}))$ the extension $\mathfrak{p}|\mathfrak{p}_1$ is unramified and

$$\delta(\mathfrak{p}) = p\,\delta(\mathfrak{p}_1) + \frac{p-1}{2} \cdot v_{\mathfrak{p}_{i+1}}(du^{p^{i+1}}) = p\,\delta(\mathfrak{p}_1) + \frac{p-1}{2} \cdot \big((r'+1)p^j + r + \ell p^{j+1}\big).$$

Putting the above together we conclude that the $\Delta$-invariants of the singular prime $\mathfrak{p}$ are given by

$$\Delta_n(\mathfrak{p}) = \begin{cases} \frac{p-1}{2} \cdot ((r+1)p^i + r'p^j + \ell p^{j+1}), & \text{if } n = 0, \\ \frac{p-1}{2} \cdot (r+1)p^{i-n}, & \text{if } 0 < n \le i-j, \\ \frac{p-1}{2} \cdot rp^{i-n}, & \text{if } i-j < n \le i, \\ 0, & \text{if } i < n. \end{cases}$$

Consequently $\delta(\mathfrak{p}) = \frac{p-1}{2} \cdot \big(rP_0^i + P_j^i + r'p^j + \ell p^{j+1}\big)$. Furthermore,

$$\deg(\mathfrak{p}_n) = \begin{cases} p^{j+1}, & \text{if } n = 0, \\ p^j, & \text{if } 1 \le n \le i-j+1, \\ p^{i+1-n}, & \text{if } i-j+1 \le n \le i+1, \\ 1, & \text{if } i+1 \le n, \end{cases}$$

and in particular the prime $\mathfrak{p}_i$ is not rational, as desired.

Both examples are essential to proving Theorem 3.5. Indeed, the two examples yield Proposition 3.7, which in turn is used to get the lower bound of $\lambda_p(d)$ in Lemma 3.8.

**Remark 4.3.** All we need in Example 4.2 for the construction of $F|K$ is a base field $K$ that is imperfect, i.e., we need a base field $K$ whose $p$-degree $\operatorname{pdeg}(K) = \log_p[K : K^p]$ is positive. By contrast, in Example 4.1 we need an imperfect base field $K$ with $p$-degree at least 2; indeed, if $\operatorname{pdeg}(K) = 1$ then $K = K^p(a)$ whenever $a \in K \setminus K^p$, i.e., $K^{1/p^n} = K^p(a^{1/p^n})$ for all $n \geq 0$ and all $a \in K \setminus K^p$, and therefore the element $b$ in (5) cannot exist. The author does not know if there exist function fields $F|K$ with $\operatorname{pdeg}(K) = 1$ that satisfy the specified properties in this example.

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